DS3001 Data Analysis & Visualization

Final Report

Sports Squad Optimizer

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I. INTRODUCTION

In today's sports world, using data to make decisions is very important for building strong teams. The Sports Squad Optimizer is a tool designed to help cricket and football coaches choose the best players for their teams. It collects detailed player information, like performance stats and injury history, and organizes it in an easy-to-use way.

This tool offers features like custom player ratings, side-by-side player comparisons, and performance tracking over time. It also uses AI to suggest the best team lineup based on recent performance and fitness. By using simple tools like Python, data analysis, and clear visuals, the optimizer helps coaches save time and make better decisions for their teams. It makes team management easier, smarter, and more effective.

Motivation

Understanding sports performance requires not just individual player data but also insights into how players interact and perform under different conditions. To achieve this, our project focuses on a rich dataset of cricket and football player statistics gathered through web scraping. This dataset includes valuable information such as player performance metrics, injury histories, and other key attributes that allow for a comprehensive analysis of player capabilities.

The dataset provides an opportunity to explore a variety of analytical tools and techniques, such as custom player ratings, trend analysis, and injury tracking. By examining this dataset, we aim to gain hands-on experience with important concepts in sports analytics, like data filtering, player comparisons, AI-based recommendations, and clustering for compatibility-based team formation. This practical approach not only enhances our understanding of sports data but also enables the development of a powerful tool to assist coaches in making smarter team selections.

Real World Relevance

The use of data-driven insights in sports management is becoming increasingly prevalent, as professional leagues and franchises recognize the competitive edge it offers. Teams in tournaments like the ICC Cricket World Cup or FIFA World Cup now rely on detailed analytics to identify strengths, weaknesses, and optimal strategies. The Sports Squad Optimizer aligns with this trend by combining statistical analysis with machine learning to address common challenges such as selecting players for specific match conditions, managing player workloads, and predicting team compatibility. By bridging the gap between raw data and actionable strategies, this project positions itself at the forefront of modern sports analytics.

Data Collection

The data used in this project was collected from Wikipedia, a widely trusted platform for detailed and comprehensive information. The data collection process was conducted using BeautifulSoup, a powerful Python library for web scraping that simplifies the extraction of information from HTML and XML pages. By leveraging BeautifulSoup, we were able to navigate through Wikipedia's player profiles and gather detailed statistics, such as match performances, batting averages, injury histories, and other relevant attributes for cricket players.

This method enabled efficient extraction of a large dataset containing various player-related attributes. The structured and diverse data collected provides a strong foundation for implementing advanced analytics and modeling, allowing us to deliver meaningful insights for team selection and performance optimization.

Data Set Description

The dataset used in this project contains detailed statistics of 572 cricket players from various teams. It provides a comprehensive view of each player's performance across different match formats (Test, ODI, First-Class, and List A) and includes data fields for their batting, fielding, and recent form. The key features in the dataset are:

- **Player Details**: Names and nationalities of the players.
- **Match Participation**: Number of matches played in different formats (Test, ODI, First-Class, List A).
- **Performance Metrics**: Runs scored, batting averages, centuries/half-centuries, top scores, and fielding stats (catches and stumpings).
- **Performance by Conditions**:
    - Home vs. Away Runs: Splits of runs scored in home versus away conditions.
    - Runs against Strong vs. Weak Teams: Performance against the top 4 and bottom 4 teams.
    - Recent Form: Runs scored in the last 2 and 5 years.
- **Advanced Metrics**: Customizable filters based on age, experience, and recent match ratings.

The dataset is well-suited for analyzing player performance trends, comparing players, and providing actionable insights for team selection. By leveraging this dataset, we implemented features like custom ratings, player comparisons, AI-based team selection, and compatibility-based clustering to assist coaches in making data-driven decisions.

```
First 5 rows of the dataset:
             Player   Nationality Test Matches ODI Matches FC Matches  \
0  Sachin Tendulkar         India          200         463        310
1       Virat Kohli         India          116         295        145
2        Brian Lara   West Indies          131         299        261
3     Ricky Ponting     Australia          168         375        289
4    Jacques Kallis  South Africa          166         329        257

   LA Matches Test Runs  ODI Runs FC Runs LA Runs  ... Test Catches/stumpings  \
0         551    15,921     18426  25,396  21,999  ...                 115/-
1         399     9,017     13906   11097   12886  ...                 113/-
2         429    11,953     10405  22,156  14,602  ...                 164/-
3         456    13,378     13704  24,150  16,363  ...                 196/-
4         424    13,289     11579  19,695  14,845  ...                 200/-

  ODI Catches/stumpings FC Catches/stumpings LA Catches/stumpings Home Runs  \
0                 140/-                186/-                175/-      9835
1                 152/-                142/-                182/-     10463
2                 120/-                320/-                177/-      9475
3                 160/-                309/-                195/-     10413
4                 131/-                264/-                162/-      2243

  Away Runs Last 5 Years Runs Last 2 Years Runs Top 4 Teams Runs  \
0      8591              8229              7431             9100
1      3443              4274               431              225
2       930              4866              1669             4238
3      3291              4780               871              410
4      9336              8717              8192             2480

  Bottom 4 Teams Runs
0                9326
1               13681
2                6167
3               13294
4                9099

[5 rows x 32 columns]
```

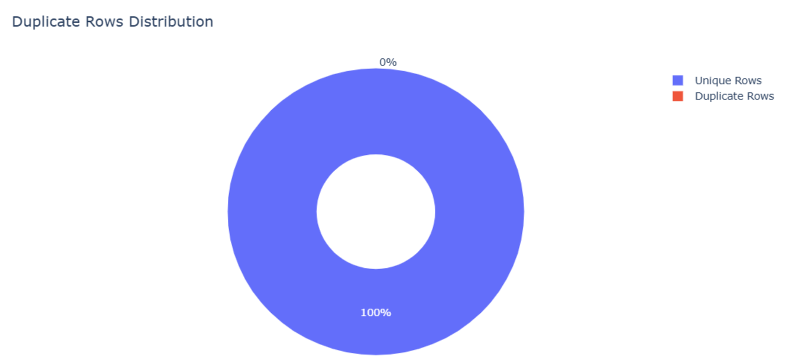
II. DATA PREPROCESSING

A. Handling Missing Values

Missing values were systematically identified across all columns and visualized using bar charts to assess the extent of the issue. Numeric columns with missing values were filled with zero, ensuring that calculations and visualizations remained accurate. Alternative imputation strategies, such as mean or median filling, were considered but not used due to the high variability in player performance metrics.

B. Duplicate Handling

Duplicate rows were identified using Pandas' `duplicated` function. Visualization via a pie chart provided a clear representation of the proportion of duplicate rows. The duplicates were removed, reducing redundancy and ensuring dataset integrity for subsequent analysis. Post-removal checks confirmed the absence of duplicates.



C. Cleaning Numeric Columns

Numeric columns with non-standard formatting (e.g., commas, dashes, and special characters) were cleaned using regular expressions. Invalid entries were converted to numeric values or replaced with zero to standardize the dataset. Specific preprocessing steps include:

- Columns like "100s/50s" and "Catches/Stumpings" were split into separate fields and normalized.
- Non-numeric characters and special symbols were removed using regex.
- Mean imputation was used to handle missing or invalid values.

These steps ensured compatibility with mathematical operations and visualizations, laying a foundation for dimensionality reduction and clustering techniques.

III. VISUALIZATION AND ANALYSIS

A. Key Visualizations

1. **Scatter Plots:**

Explored relationships between "ODI Matches" and "ODI Runs," revealing player performance trends. The inclusion of player-specific colors enhanced interpretability:

Plotted "ODI Runs" versus "ODI Averages" across formats to identify consistent performers and outliers.

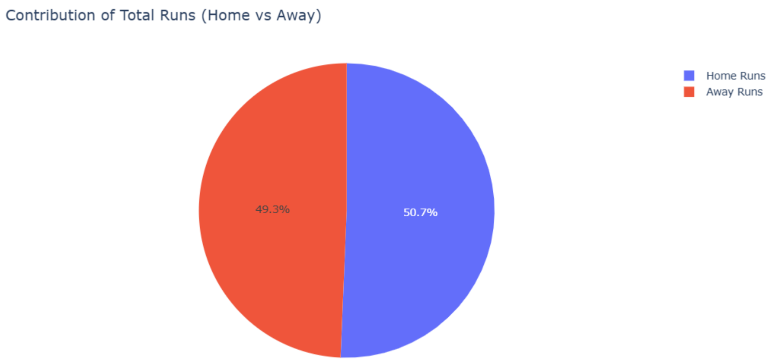
2. Bar Charts:

Displayed the distribution of matches across formats, highlighting players' participation in different formats:

Visualized century contributions across formats to emphasize players' scoring abilities.

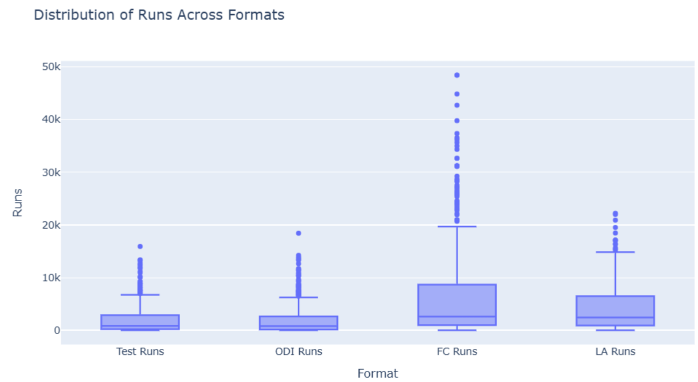
3. **Pie Charts:**

Compared home and away runs, providing insights into players' adaptability to different playing conditions:



4. **Box Plots:**

Highlighted the spread and outliers in runs scored across formats. This analysis was useful for identifying players with extreme performance variations:

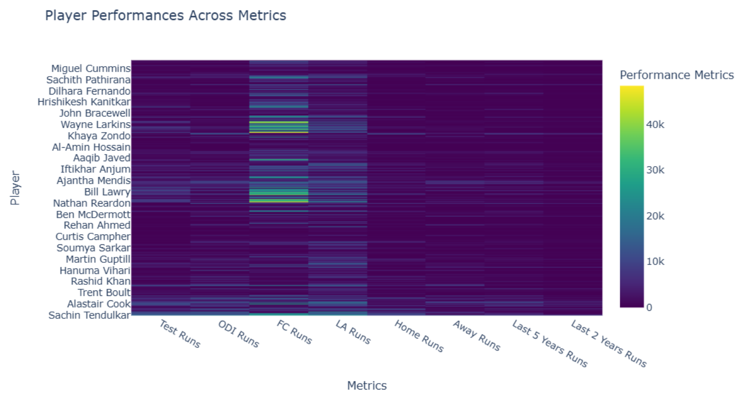


5. **Radar Charts:**

Visualized individual player strengths across formats, facilitating easy comparison of their contributions in Test, ODI, FC, and LA matches.

6. **Heatmaps:**

Aggregated performance metrics for all players, showing strengths and weaknesses across multiple metrics at a glance:



7. **Treemaps:**

Illustrated the distribution of centuries scored across formats, offering a hierarchical view of players' contributions to team success:

8. **t-SNE Visualization**:

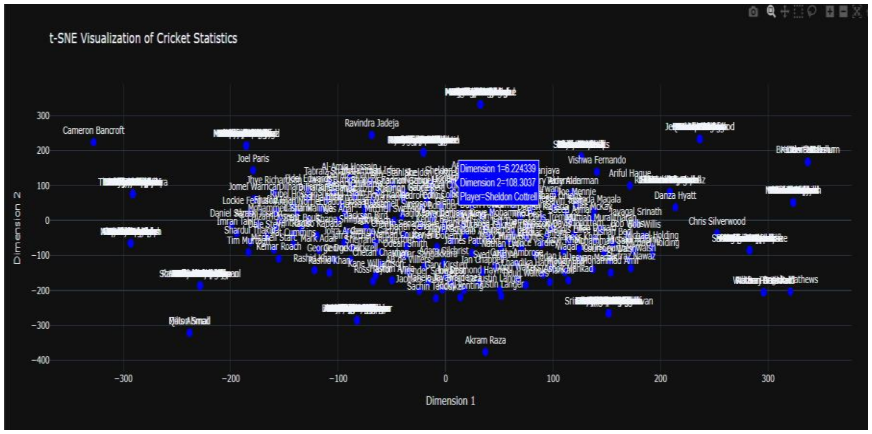
Applied t-SNE to reduce the dimensionality of the dataset, enabling a clear 2D visualization of player compatibility.

A scatterplot visualized player clusters, aiding in compatibility-based team selection.

9. **Clustering Techniques**:

KMeans clustering was applied to group players into compatible teams based on proximity in t-SNE reduced dimensions.

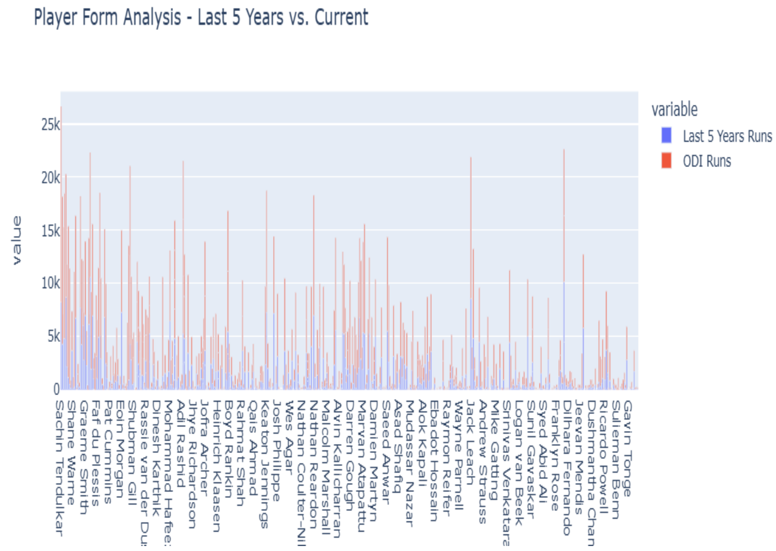
DBSCAN offered an alternative clustering approach, identifying natural groupings without predefining the number of clusters.



B. Derived Metrics

Derived metrics played a significant role in uncovering key insights into player performances. The following metrics were calculated and analyzed:

1. **Last 5 Years Runs vs. Last 2 Years Runs**:
    a. These metrics provided a view of recent form and consistency. Players with increasing runs over the last two years were identified as in-form contributors, while those with decreasing performance highlighted potential areas for intervention.
    b. Below is a comparative visualization of average runs in the last 5 and 2 years:

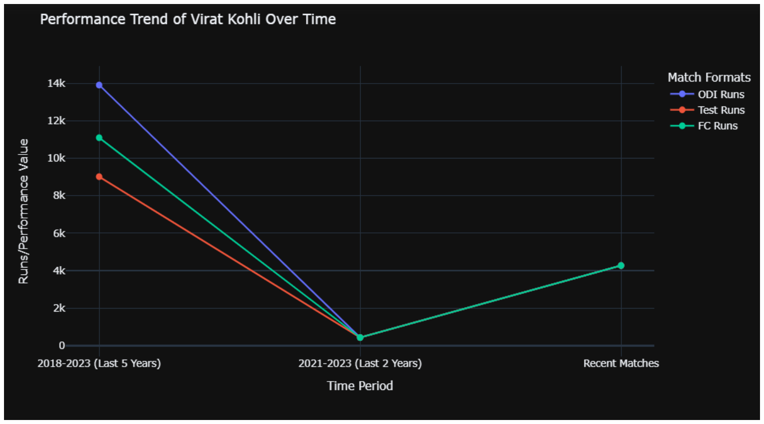


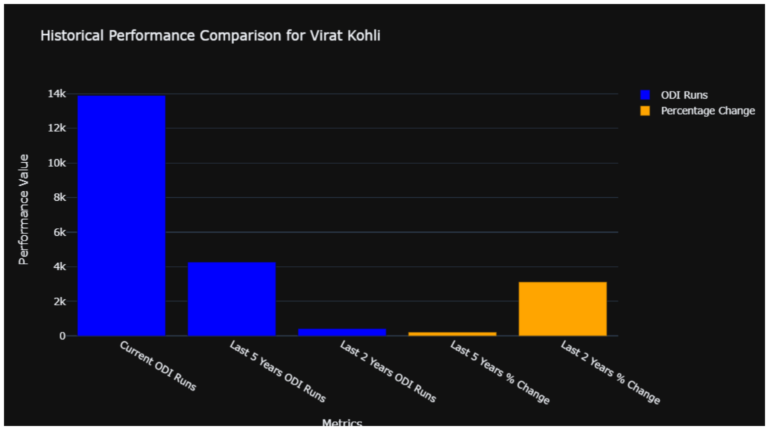
2. **Home vs. Away Performance**:
    a. A detailed comparison revealed players who excelled in home conditions but struggled away. This insight was critical for selecting players for international tours.

3. **Format-Based Metrics**:
    a. Aggregated metrics such as "Average Runs" and "Total Centuries" across Test, ODI, and

FC formats enabled cross-format comparisons. This analysis identified format specialists and all-round performers.

4. **Performance Consistency**:
    a. Standard deviations and coefficient of variation (CV) were calculated for key metrics like runs and averages. Players with low CV were highlighted as consistent performers, while those with high CV showed variable performances.
5. **Recent Form vs. Career Average**:
    a. The difference between recent form (last 2 years) and career averages highlighted players whose current performance deviated significantly from their career benchmarks. This allowed identification of rising stars and players past their prime.





IV. MACHINE LEARNING AND TEAM OPTIMIZATION

A. Machine Learning Models

- To evaluate player selection and performance, a variety of machine learning approaches were applied:
- **Linear Regression**:
    - Predicted player selection likelihood as a continuous variable.
    - Accuracy was measured by rounding predictions to binary outputs.
- **Logistic Regression**:
    - Modeled player selection probability based on performance metrics. This approach offered a clear interpretation of feature importance.
- **Support Vector Machine (SVM)**:
    - Classified players using hyperplanes optimized for maximum separation. Despite its effectiveness in high-dimensional spaces, it required careful tuning for balanced performance.
- **Decision Tree Classifier**:
    - Provided a visual and interpretable method for classifying players based on hierarchical rules derived from performance metrics.

B. Justification of the Models

The choice of machine learning models was guided by their ability to handle the complexity and nature of the dataset. Linear regression provided a baseline for understanding feature relationships, while logistic regression offered interpretable probabilities of player selection. The support vector machine (SVM) and decision tree classifier were chosen for their ability to model non-linear relationships, with SVM excelling in high-dimensional spaces. Finally, the artificial neural network (ANN) was selected for its superior capability in capturing complex interactions among features, making it the most effective model for team optimization. Each model was rigorously evaluated, ensuring that the final recommendations were both accurate and reliable.

B. Deep Learning Approach

- **Neural Network for Team Selection**: A feedforward neural network was implemented to optimize team selection. The model evaluated player combinations based on their aggregated performance metrics to recommend the best squad for specific match conditions.
- **ANN Architecture**:
  - Input Layer: Processed normalized player features.
  - Hidden Layers: Two dense layers with 16 and 8 neurons, using ReLU activation for non-linearity.
  - Output Layer: A single neuron with a sigmoid activation function for binary classification.
- **Training Process**:
  - The ANN model was trained over 50 epochs with a batch size of 8. A validation split of 20% was used to monitor generalization.
  - Below is the visualization of the training process:

C. Compatibility-Based Team Formation

- **Distance-Based Teaming**:
  - Euclidean distance was calculated to identify the closest players, forming balanced teams of 11 members.
  - Ensured compatibility by considering statistical proximity.
- **Clustering**:
  - KMeans and DBSCAN segmented players into naturally compatible groups, emphasizing balanced team composition.

D. Results

- Accuracy Scores:

  Linear Regression: Moderate accuracy, dependent on feature scaling.

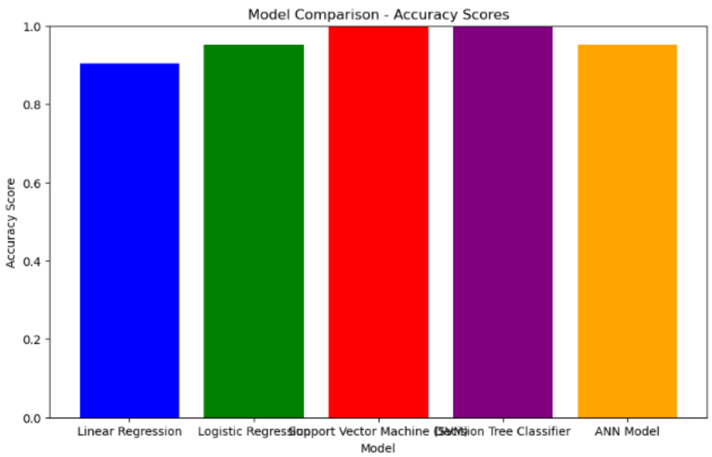
  Logistic Regression: Provided reliable predictions with interpretable weights.

  SVM: High accuracy but computationally expensive.

  Decision Tree: Easy to interpret but prone to overfitting.

  ANN Model: Achieved the highest accuracy with strong generalization capabilities.

  Below is a comparison of accuracy scores across models:



The ANN model demonstrated superior performance in recommending optimal squads, with clear advantages in handling complex, nonlinear interactions among player metrics.

V. INSIGHTS

The analysis provides actionable recommendations for coaches and team managers. For instance, players identified as format specialists can be prioritized for matches where their strengths align with team needs, while versatile players with balanced home and away performances can be critical for international tours. Additionally, metrics like "Recent Form vs. Career Average" enable targeted interventions, such as focusing training efforts on players with declining performance or providing opportunities to rising stars. These insights ensure that teams are not only optimized for current performance but also poised for long-term success.

Consistent high-scorers were identified as key assets for team strategy. Players with balanced performance across home and away matches were highlighted as versatile contributors.

Outlier analysis revealed players with exceptional performance in specific formats, suggesting potential specialization roles.

```
Player of the Week: Jacques Kallis
Top Performers:
                Player  Last 2 Years Runs
4       Jacques Kallis               8192
0     Sachin Tendulkar               7431
20        Graeme Smith               6911
26         Ross Taylor               6621
16         Chris Gayle               6450
12      Adam Gilchrist               6193
372         Rory Burns               5535
256        Steve Waugh               5130
23         Hashim Amla               4794
92     Mohammad Hafeez               4644
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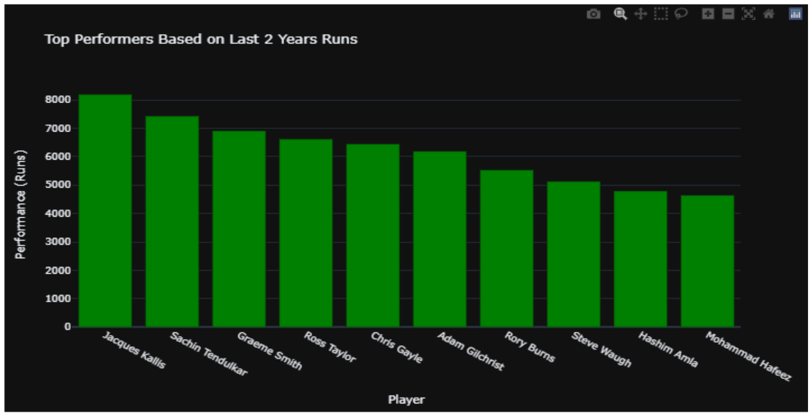

VI. CONCLUSION

The analysis highlighted the importance of structured preprocessing, advanced visualization techniques, and machine learning approaches in analyzing cricket performance data. Key findings underscored the variability in player contributions across formats and conditions.

Dimensionality reduction through t-SNE and clustering methods (KMeans and DBSCAN) enhanced compatibility-based team formation. The integration of deep learning methods showcased the utility of AI in sports analytics, underscoring its role in improving team management strategies.

VII. FUTURE DIRECTIONS

While the Sports Squad Optimizer provides a robust framework for team analysis, there is significant potential for future enhancements. Incorporating real-time data streams from live matches or integrating physiological metrics like fitness levels and recovery times could further refine the recommendations. Expanding the tool to include other sports or developing a mobile-friendly application interface would increase its accessibility and utility. Additionally, applying advanced AI techniques, such as reinforcement learning, could enable dynamic strategy adjustments during live games, transforming how teams adapt to changing match conditions.